PROCRASTINATION PREDICTOR

(Project Report)

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PROJECT OVERVIEW :-

In student life, many factors affect performance such as study time, sleep, stress, and distractions like mobile usage. At the same time, procrastination is a common problem where students delay their work.

This project is developed to predict two things:

- Student performance (marks)
- Procrastination risk

The main idea is to use basic machine learning to understand how daily habits affect results and then give useful suggestions.

OBJECTIVE :-

The objectives of this project are:

- To understand the basics of machine learning
- To build a prediction model from scratch using Python
- To analyze student habits and their impact on performance
- To predict procrastination risk
- To give simple suggestions based on the result

METHODOLOGY :-

The project follows a simple step-by-step approach.

First, a dataset is created manually. It includes details like study hours, sleep, screen time, attendance, assignments, and stress level. It also includes marks and whether the student procrastinated or not.

Next, the data is normalized. This means all values are converted into a similar range so that the model can learn properly.

Then, two models are trained:

- One model predicts marks using a regression approach
- The second model predicts procrastination risk using probability

After training, the user enters their own data. The program then calculates:

- Predicted marks
- Grade
- Procrastination risk

Finally, based on these results, the system gives suggestions.

FEATURES USED :-

The model uses the following inputs:

- Sleep hours
- Screen time
- Study hours
- Number of assignments
- Attendance percentage
- Social media usage
- Stress level

These factors are important because they directly affect productivity and focus.

WORKING OF THE MODEL :-

The model works in the following way:

- It calculates a weighted sum of input values
- For marks prediction, a linear model is used

- For procrastination, a sigmoid function is used to get probability

  If the probability is high, it means the student is likely to procrastinate.

  The marks are also used to assign a grade like A, B, C, or D.

RESULT AND EXAMPLE :-

The system gives the following output:

- Predicted marks (0–100)
- Grade (A, B, C, D)
- Procrastination risk (in percentage)
- A simple suggestion for improvement

```
Study hours: 8
Sleep hours: 6
Screen time: 4
Attendance (%): 90
Assignments completed: 2
Stress level (1-10): 5

Predicted Marks: 83.75
Grade: B
Procrastination Risk: 60.86%
You may procrastinate. Try better time management.
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CONCLUSION :-

This project helped me understand the basics of machine learning and how models are trained from scratch. It also shows how daily habits can be analyzed using data.

Although the dataset is small, the concept can be expanded with more data to improve accuracy.

FUTURE SCOPE :-

- Use a larger and real dataset
- Add more features like mood or deadlines
- Improve accuracy using advanced algorithms
- Convert this into a mobile or web application